

Table 1: Generalized methodology framework for phased capability transition in the European Cloud-Edge-IoT Cybersecurity Roadmap. Each row represents a distinct maturity stage, assessed across four analytical dimensions: R&I maturity (measured via the Operational Deployment Phase, ODP, scale), the policy and standards environment, the operational deployment reality, and the market and sovereignty position (measured via the Strategic Sovereignty Level, SSL, scale). Gate rows define the minimum verifiable conditions required for a transition to be declared. The table is reused across all ten strategic priorities; priority-specific progression matrices instantiate this framework with domain-specific evidence.

Phase / Strategic Sovereignty Level (SSL) Band	R&I Maturity measured via ODP: Applied R&D → Fragmented Piloting → Standardised Integration → Ubiquitous Operation	Policy, Regulatory & Standards governance formation · compliance obligation · certification maturity · standards formalisation	Operational Deployment & Adoption deployment scale · actor types · integration depth · sector coverage	Market Uptake & Strategic Sovereignty measured via SSL: High Foreign Dependency → Emerging EU Ecosystem → Full Sovereign Autonomy
I. Current Status <i>Baseline diagnosis</i> SSL: High Foreign Dependency	What the baseline shows in research and innovation This cell captures the present level of scientific maturity, prototype availability, and technical readiness. It should summarise what already exists, what is still experimental, and where major capability gaps remain. Evidence may include low-TRL results, fragmented pilots, isolated demonstrators, or dependence on external technologies. <i>Methodological role: establishes the starting maturity level against which later technical progress is assessed.</i>	What the baseline shows in policy and standards This cell describes the current regulatory and governance environment. It should indicate whether relevant legislation is absent, emerging, partially enforced, or already shaping behaviour. It may also note missing standards, immature certification schemes, or inconsistent national implementation. <i>Methodological role: defines the initial institutional conditions that constrain or enable capability development.</i>	What the baseline shows in deployment and adoption This cell explains how far the capability has progressed in practice. It should state whether deployment is absent, limited to pilots, confined to research settings, or used by a narrow group of early adopters. It also identifies the main operational bottlenecks. <i>Methodological role: anchors the real-world starting point, not only the laboratory or policy position.</i>	What the baseline shows in market and strategic position This cell characterises the current competitive and sovereignty situation. It should indicate external dependency, weak industrial presence, lack of procurement pull, limited SME participation, or structural asymmetry versus dominant foreign actors. <i>Methodological role: establishes the initial strategic exposure and market gap that the roadmap aims to reduce.</i>
<i>Gate I → II</i>	This gate defines the minimum conditions required to move from baseline diagnosis to Phase 1 action. It should contain named and verifiable triggers across all four dimensions: a research trigger (e.g. launch of targeted calls or validated early prototypes), a policy trigger (e.g. first guidance, implementing acts, or initial governance framework), an operational trigger (e.g. pilot initiation), and a market trigger (e.g. first investment signal, funding instrument, or procurement incentive).			

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Phase / SSL Band	R&I Maturity (ODP)	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Sovereignty (SSL)
II. Phase 1 <i>Pilot Validation & Applied Maturity</i> SSL: Early EU Capability Emergence	What Phase 1 means for research and innovation This cell describes the transition from fragmented research to applied validation. It should cover targeted R&I activity, prototype consolidation, and movement from exploratory work toward demonstrable use conditions. Typical evidence includes rising TRLs, benchmarking, and early integration into priority use cases. Transition criterion: validated prototype evidence or TRL 4–5 maturity threshold. <i>Methodological role: shows that technical ideas are becoming credible, structured, and testable under realistic conditions.</i>	What Phase 1 means for policy and standards This cell captures the first conversion of political intent into usable regulatory or standards-based direction. It may include guidance documents, implementing rules, pre-standardisation work, or the first compliance frameworks that create clear expectations for adopters. Transition criterion: initial regulatory guidance, pre-standardisation output, or first compliance framework. <i>Methodological role: shows how the policy environment begins to create directional pull and reduce uncertainty.</i>	What Phase 1 means for deployment and adoption This cell covers pilot deployment, first operational testing, and limited adoption by selected actors or sectors. It should explain where the capability is being trialled, under what constraints, and what kinds of users or infrastructures are engaged. Transition criterion: at least one structured pilot deployment in a representative environment. <i>Methodological role: demonstrates that the capability is no longer purely conceptual and is entering applied operational contexts.</i>	What Phase 1 means for market and strategic position This cell explains whether the first signs of industrial traction appear. It may include early procurement references, investment trajectories, SME support mechanisms, or the emergence of a business case linked to compliance or sovereignty needs. Transition criterion: first investment signal, procurement reference, or industrial interest. <i>Methodological role: indicates whether experimentation is beginning to translate into strategic and economic relevance.</i>
<i>Gate II → III</i>	<i>This gate marks the shift from validated pilots to broader scaling. It should specify evidence that Phase 1 results are stable enough to support industrialisation or structured expansion. Triggers usually include validated prototypes, operational policy instruments, successful pilots in multiple settings, and visible market commitment or investment momentum.</i>			
III. Phase 2 <i>Industrial Scaling, Standardisation & Cross-Sector Deployment</i> SSL: Emerging Autonomy / Structured Sovereignty	What Phase 2 means for research and innovation This cell describes the coexistence of advanced applied R&I and industrial consolidation. It should show how research feeds productisation, performance optimisation, and architecture refinement. Evidence may include volume-ready designs, mature reference implementations, or cross-domain integration. Transition criterion: reproducible reference implementation and validated scale-up readiness. <i>Methodological role: demonstrates that innovation is no longer isolated, but becomes a reproducible capability base.</i>	What Phase 2 means for policy and standards This cell captures the formalisation stage. It should reflect ratified standards, sector rules, common compliance profiles, or interoperable certification pathways that transform pilots into repeatable practices. This is where voluntary direction becomes structured governance. Transition criterion: ratified standards, formal policy instruments, or interoperable certification pathway. <i>Methodological role: explains how institutional coherence enables repeatability, interoperability, and trust at scale.</i>	What Phase 2 means for deployment and adoption This cell explains cross-sector rollout and operational scaling. It should show how deployments extend beyond isolated pilots into multiple sectors, jurisdictions, or infrastructures, with improving reliability and integration into business or public-service operations. Transition criterion: multi-sector deployment beyond isolated pilots. <i>Methodological role: marks the transition from pilot relevance to system-level operational presence.</i>	What Phase 2 means for market and strategic position This cell shows whether a meaningful industrial and strategic position is being built. It may cover rising market share, growing procurement penetration, decreasing foreign dependency, and the emergence of continental-scale actors or federated offers. Transition criterion: measurable market uptake and reduced strategic dependency. <i>Methodological role: assesses whether Europe is moving from technology participation toward structured competitive capacity.</i>
<i>Gate III → IV</i>	<i>This gate defines readiness for sovereign leadership. It should identify when the capability is sufficiently mature, standardised, deployed, and commercially credible to support autonomous large-scale operation. Triggers usually include mature products or infrastructures, binding standards or rules, continental deployment evidence, and proof of competitive viability.</i>			

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Phase / SSL Band	R&I Maturity (ODP)	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Sovereignty (SSL)
IV. Phase 3 <i>Autonomous Leadership & Frontier Innovation</i> SSL: Full Sovereign Autonomy	What Phase 3 means for research and innovation This cell captures leadership-level innovation. It should describe frontier R&I, continuous optimisation, and the ability to define new technical paradigms rather than merely adopting them. Evidence may include world-leading reference architectures, breakthrough performance gains, or agenda-setting research. Transition criterion: frontier-level innovation capability and autonomous optimisation. <i>Methodological role: shows that the capability has moved from catch-up to leadership and future-shaping innovation.</i>	What Phase 3 means for policy and standards This cell describes regulatory and standards leadership. It should show that the relevant frameworks are coherent, enforceable, and influential beyond the original jurisdiction, possibly shaping international practice. Transition criterion: stable regulatory leadership and standards-setting influence. <i>Methodological role: indicates that governance has shifted from reactive adaptation to rule-setting capacity.</i>	What Phase 3 means for deployment and adoption This cell explains full operationalisation. It should describe routine, large-scale, resilient deployment across infrastructures and sectors, with the capability treated as a default operating baseline rather than a special programme or exception. Transition criterion: full-scale operationalisation across infrastructures and sectors. <i>Methodological role: confirms that the capability is embedded in the normal functioning of the ecosystem.</i>	What Phase 3 means for market and strategic position This cell captures competitive parity or leadership. It should indicate a strong industrial base, reduced structural dependency, export potential, and strategic influence over ecosystem evolution. Transition criterion: durable competitive parity or strategic leadership. <i>Methodological role: assesses whether capability development has translated into durable strategic advantage.</i>
Gate IV → V	<i>This gate marks the transition from autonomous capability to the intended end-state vision. It should summarise the evidence that the capability is not only deployed and sovereign, but durable, exportable, and structurally embedded. Triggers often include recognised leadership, stable policy coherence, continent-wide operational normalisation, and positive strategic spillovers.</i>			
V. Strategic End-State Objective <i>Long-term Vision</i> SSL: Full Autonomy + Strategic Influence	What the end-state means for research and innovation This cell describes the intended final technical condition of the roadmap. It should present the capability as mature, self-sustaining, and able to generate further innovation internally. Research no longer serves only catch-up, but continuously renews leadership. <i>Methodological role: defines the target state against which all preceding technical progression is interpreted.</i>	What the end-state means for policy and standards This cell explains the desired governance condition at maturity. It should show a coherent and stable framework in which regulation, certification, and standards operate as an integrated system that supports trust and predictability. <i>Methodological role: defines the target institutional configuration for long-term capability sustainability.</i>	What the end-state means for deployment and adoption This cell presents the operational vision once the capability has become normal, reliable, and widely embedded. It should express that deployment is no longer an exception or a programme milestone, but the accepted baseline across the target ecosystem. <i>Methodological role: defines the operational destination of the transition pathway.</i>	What the end-state means for market and strategic position This cell captures the long-term strategic ambition. It should show that the capability supports competitiveness, autonomy, resilience, and wider influence, with sustainable participation across large actors, public institutions, and SMEs. <i>Methodological role: defines the final strategic outcome that justifies the phased roadmap as a whole.</i>